# Handwriting Model

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# 1 KNN Train Test Split

## 2 Explaining Folds

* Remember in 100% of data, when do train split, it's usually 70% data for training and 30% testing.

In that 70% of training data:

```
from sklearn.model_selection import KFold
folds = KFold(n_splits = 5, shuffle = true, random_state = 42)
```

How it works?

- Let's say we have 700 pictures. We divide these into 140 pictures in each chunk.
- So we have 1,2,3,4,5 chunk.
- let k (not n splits) is 3, in this iteration, 1,2,3,4,5 is randomized
- For first round (1), 1,2,3,4 is Fold used for training, and 5 is used for validation
- Goes on until five rounds, validation data helps choose K

| Round | Train on | Validate on |
|---|---|---|
| 1 | Folds 2, 3, 4, 5 | Fold 1 |
| 2 | Folds 1, 3, 4, 5 | Fold 2 |
| 3 | Folds 1, 2, 4, 5 | Fold 3 |
| 4 | Folds 1, 2, 3, 5 | Fold 4 |
| 5 | Folds 1, 2, 3, 4 | Fold 5 |

**Note: This is only setting the instructions of loading the data for one k-value!

## 3 Explaining GridSearch CV

```
from sklearn.model_selection import GridSearchCV
optimal_k = GridSearchCV(
    estimator=KNeighborsClassifier(),
    param_grid={"n_neighbors": range(1, 30)},
    cv=folds,
)
```

- The KFold is where each k is tested.
- It runs each k, and each k runs KFold.
- Repeats until each iteration is done.
- Note that the code above hasn't load the data yet, it's just setting the instructions